Data and Artificial Intelligence

Cyber Shujaa Program

Week 2 Assignment
Data Wrangling with Netflix Dataset(Kaggle)

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1. Introduction

This assignment focused on developing hands-on experience in data wrangling using Pandas with a dataset obtained from Kaggle: [Netflix Movies and TV Shows]().
The task provided an opportunity to practice loading, exploring, cleaning, structuring, and validating data before exporting it into a reusable format.

2. Objectives

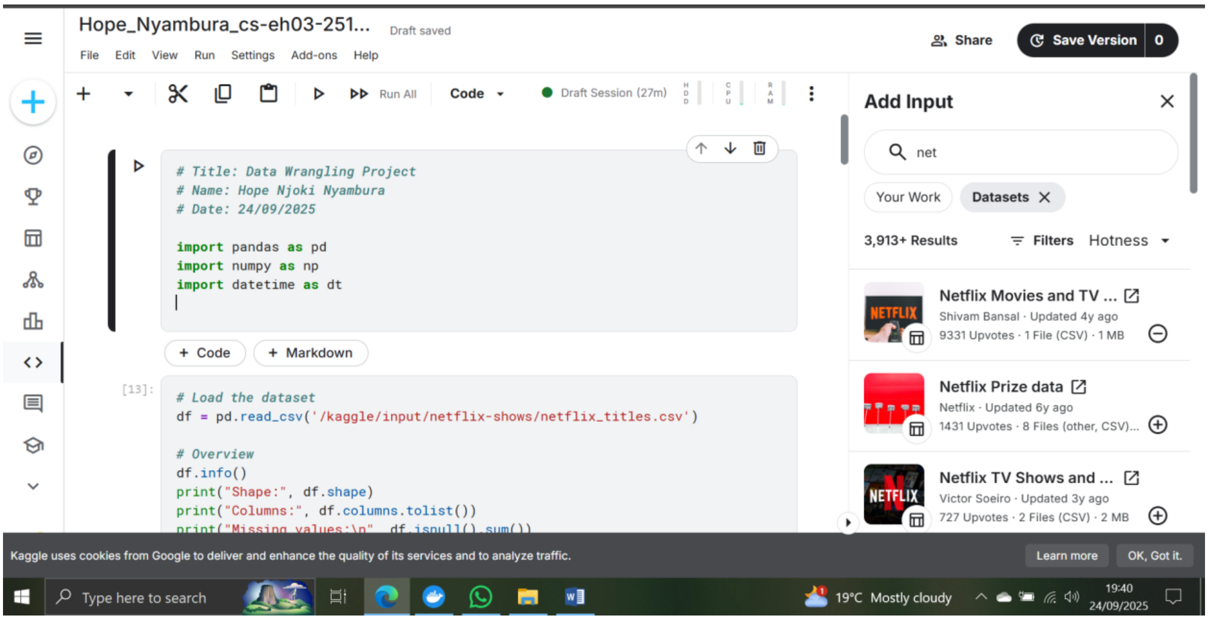
The main objectives of this assignment were:

- To load and explore the Netflix dataset using Pandas.
- To assess data types, missing values, and potential quality issues.
- To clean the dataset by handling duplicates, missing values, and formatting inconsistencies.
- To transform and enrich the dataset e.g. splitting duration into numeric and unit.
- To validate the final dataset by ensuring consistency, completeness, and logical accuracy.
- To export the cleaned dataset to CSV for future analysis or visualization.

3. Tasks Completed

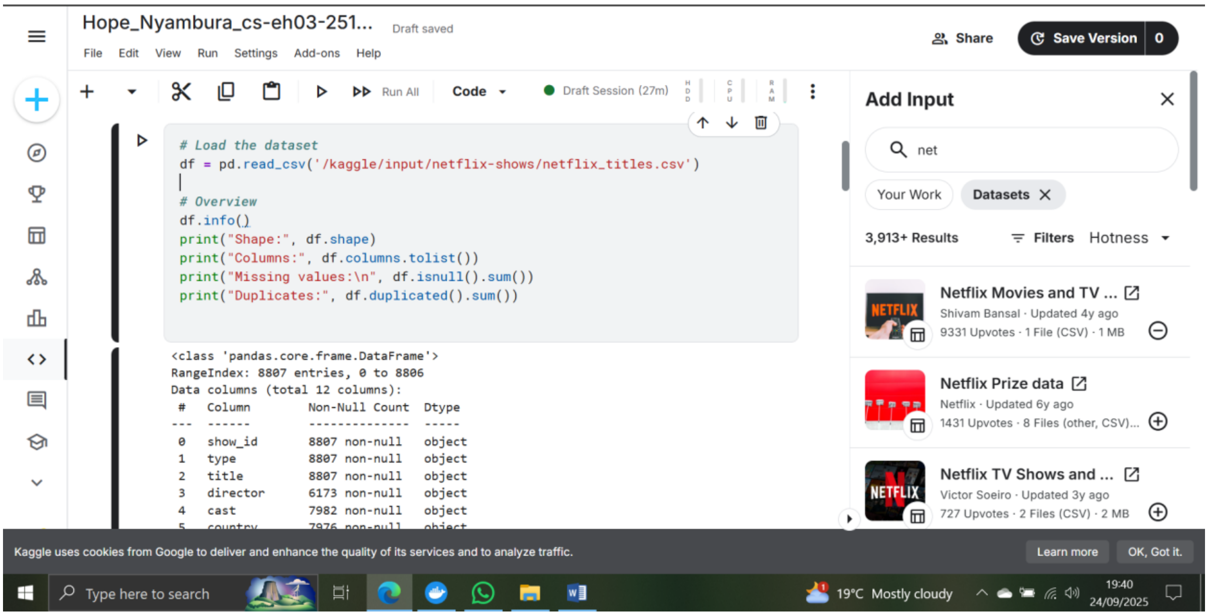
Step 1: Import Libraries

Imported Pandas, Numpy, and Datetime libraries.



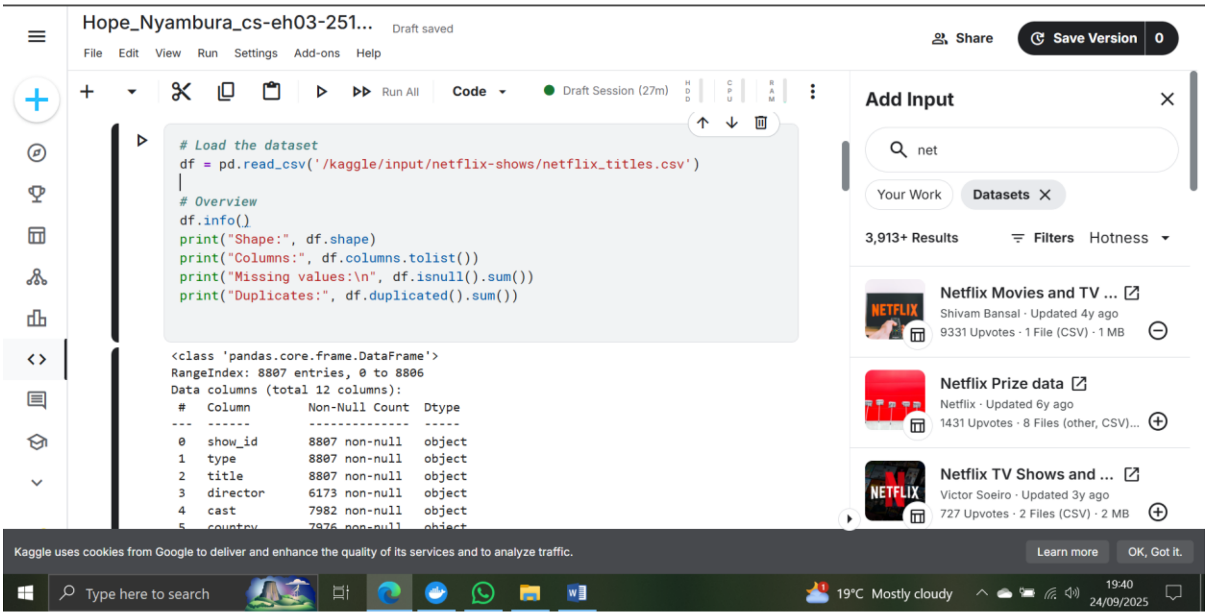
Step 2: Load Dataset

Loaded the Netflix dataset from Kaggle into a Pandas DataFrame.



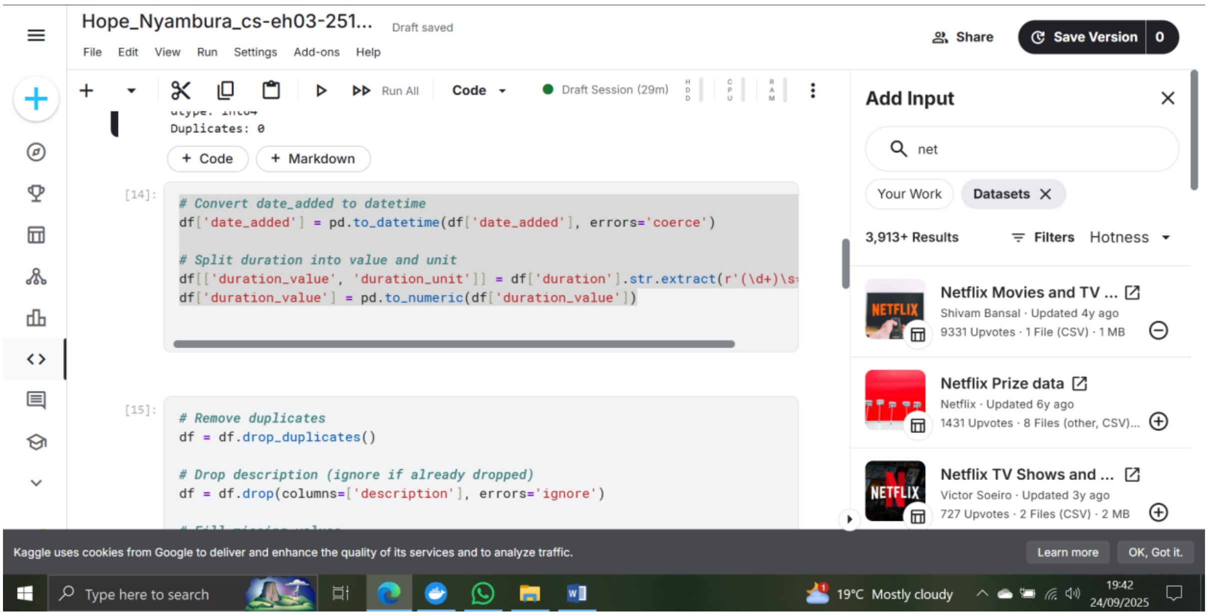
Step 3: Data Discovery

- Checked number of rows and columns.
- Inspected column names and data types.
- Assessed missing values and duplicate rows.



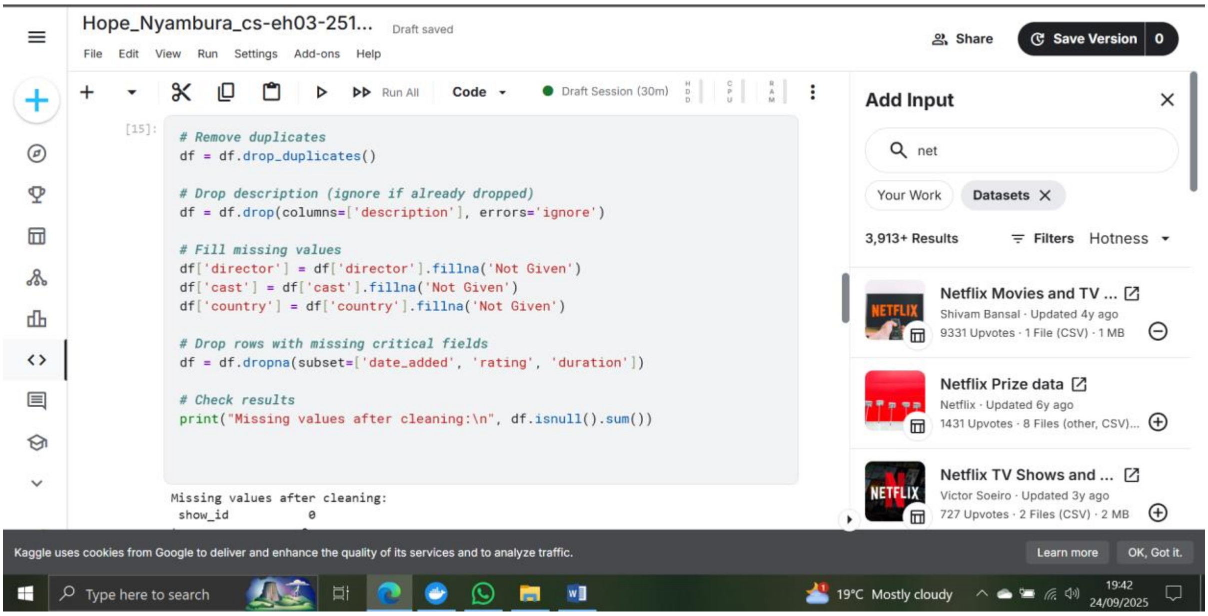
Step 4: Structuring the Data

- Converted date_added column to datetime.
- Split duration into duration_value and duration_unit.



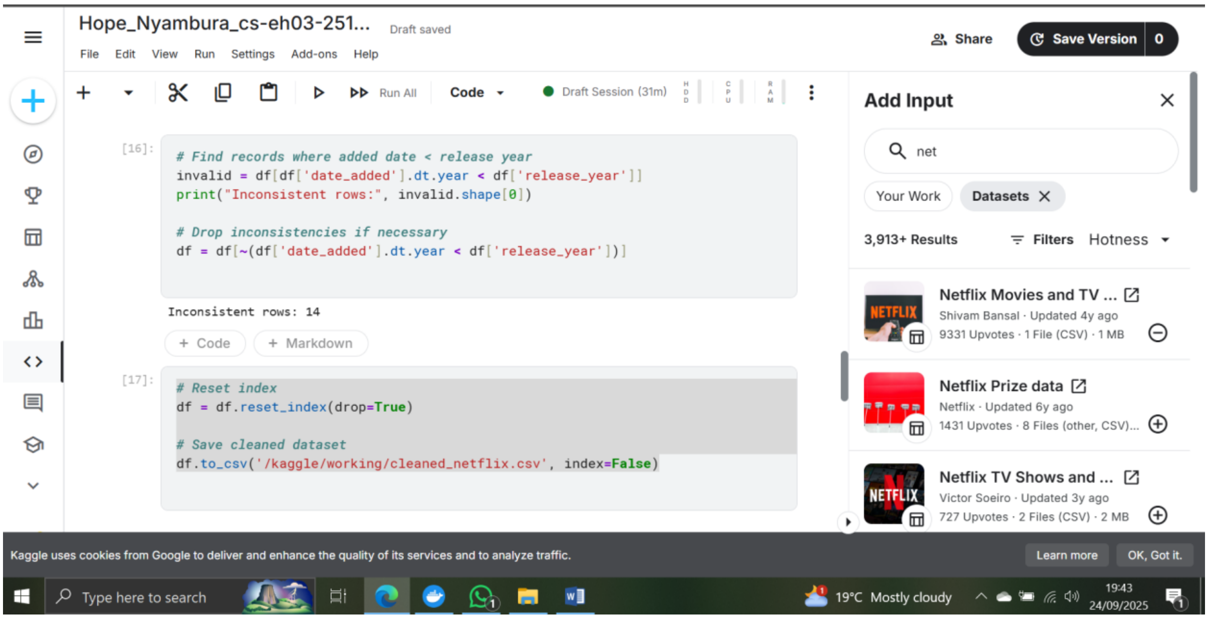
Step 5: Cleaning the Data

- Removed duplicates.
- Dropped description column.
- Filled missing values (director, cast, country) with "Not Given".
- Dropped rows missing critical values (date_added, rating, duration).



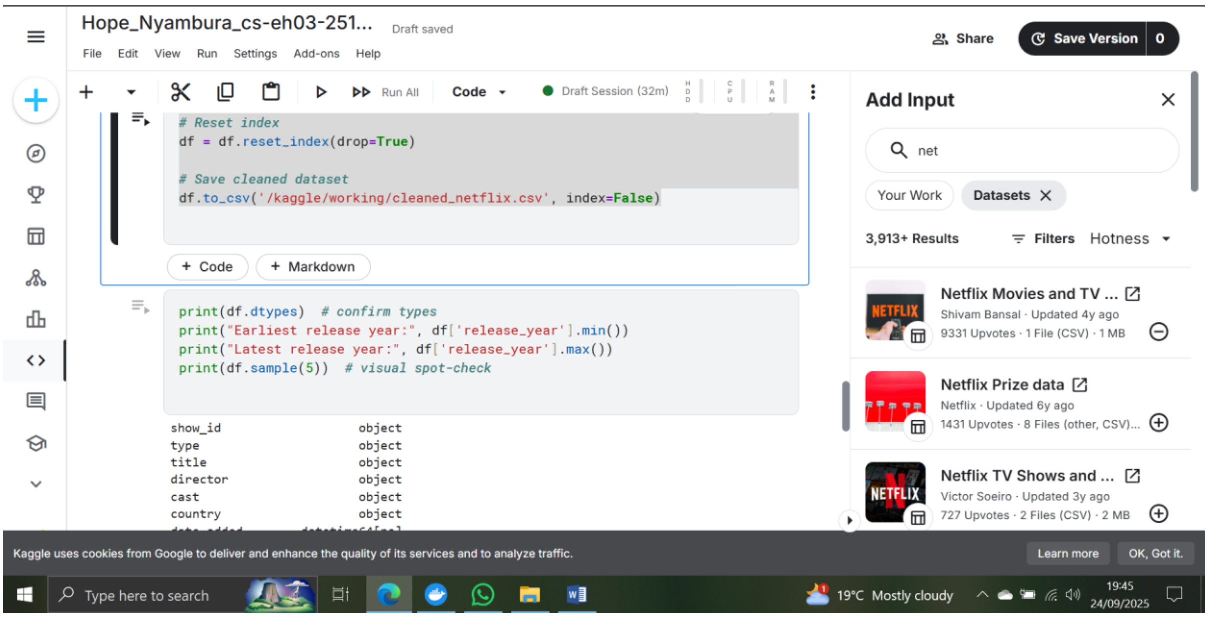
Step 6: Validation

- Ensured date_added is datetime and duration_value is numeric.
- Checked for inconsistencies where date_added < release_year and removed 14 rows.
- Confirmed earliest release year is 1925 and latest is 2021**.**
- Randomly sampled rows to visually confirm cleaning accuracy.



Step 7: Export Final Dataset

- Reset DataFrame index.
- Exported cleaned dataset to CSV: cleaned_netflix.csv.



4. Conclusion

This assignment provided valuable practice in end-to-end data wrangling using Pandas. I successfully cleaned and structured the Netflix dataset, resolved missing values, and validated the results. The cleaned dataset is now consistent, complete, and ready for further analysis or visualization.

5. Link to Kaggle Notebook

[https://www.kaggle.com/code/defhopeee/hope-nyambura-cs-eh03-25100]